

1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

47	13.721658700	fe80::6962:fd55:6f5...	fe80::6231:92ff:fe0...	DNS	98	Standard query 0xc6a A bx-ring.msedge.net
48	13.722006500	fe80::6962:fd55:6f5...	fe80::6231:92ff:fe0...	DNS	98	Standard query 0xcf58 AAAA bx-ring.msedge.net
49	13.726058100	fe80::6231:92ff:fe0...	fe80::6962:fd55:6f5...	DNS	184	Standard query response 0xc6a A bx-ring.msedge.net CNAME bx-ring.bx-9999.bx-msedge.net CNA
50	13.728771300	fe80::6231:92ff:fe0...	fe80::6962:fd55:6f5...	DNS	208	Standard query response 0xcf58 AAAA bx-ring.msedge.net CNAME bx-ring.bx-9999.bx-msedge.net

47	13.721658700	fe80::6962:fd55:6f5...	fe80::6231:92ff:fe0...	DNS	98	Standard query 0xc6a A bx-ring.msedge.net
48	13.722006500	fe80::6962:fd55:6f5...	fe80::6231:92ff:fe0...	DNS	98	Standard query 0xcf58 AAAA bx-ring.msedge.net
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50	13.728771300	fe80::6231:92ff:fe0...	fe80::6962:fd55:6f5...	DNS	208	Standard query response 0xcf58 AAAA bx-ring.msedge.net CNAME bx-ring.bx-9999.bx-msedge.net

20	4.910740700	2402:a00:408:5605:1...	2001:4860:4829:7700...	QUIC	169	Protected Payload (KP0), DCID=e6d17cc9c8496680
21	4.911854500	2001:4860:4829:7700...	2402:a00:408:5605:1...	QUIC	695	Protected Payload (KP0)
22	4.911856400	2001:4860:4829:7700...	2402:a00:408:5605:1...	QUIC	454	Protected Payload (KP0)
23	4.912147600	2402:a00:408:5605:1...	2001:4860:4829:7700...	QUIC	93	Protected Payload (KP0), DCID=e6d17cc9c8496680
24	4.924082800	2001:4860:4829:7700...	2402:a00:408:5605:1...	QUIC	188	Protected Payload (KP0)
25	4.924538400	2402:a00:408:5605:1...	2001:4860:4829:7700...	QUIC	93	Protected Payload (KP0), DCID=e6d17cc9c8496680
26	4.952714500	2001:4860:4829:7700...	2402:a00:408:5605:1...	QUIC	88	Protected Payload (KP0)

2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

```
C:\Windows\system32>ping zomato.com

Pinging zomato.com [52.220.88.101] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 52.220.88.101:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\Windows\system32>ping spotify.com

Pinging spotify.com [2600:1901:1:7c5::] with 32 bytes of data:
Reply from 2600:1901:1:7c5::: time=13ms
Reply from 2600:1901:1:7c5::: time=13ms
Reply from 2600:1901:1:7c5::: time=13ms
Reply from 2600:1901:1:7c5::: time=13ms

Ping statistics for 2600:1901:1:7c5:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 13ms, Average = 13ms
```

- Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Apps (5)						
> Google Chrome (14)	32.4%	544.8 MB	0.1 MB/s	0 Mbps	Very high	Moderate
> Microsoft Word (32 bit) (2)	0%	116.3 MB	0 MB/s	0 Mbps	Very low	Very low
> Snipping Tool	0%	2.8 MB	0 MB/s	0 Mbps	Very low	Very low
> Task Manager	6.9%	20.7 MB	0.1 MB/s	0 Mbps	Low	Very low
> Windows Command Processor	0%	0.8 MB	0 MB/s	0 Mbps	Very low	Very low

- Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

	192.168.1.14	1980 ms	[n/a]	[n/a]	[n/a]	01:00:5E:1F:3...	[n/a]	80,443,8080
	192.168.1.15	[n/a]	[n/s]	[n/s]	[n/s]	[n/s]	[n/s]	[n/s]
	192.168.1.16	[n/a]	[n/s]	[n/s]	[n/s]	[n/s]	[n/s]	[n/s]
	192.168.1.17	[n/a]	[n/s]	[n/s]	[n/s]	[n/s]	[n/s]	[n/s]
	192.168.1.18	[n/a]	[n/s]	[n/s]	[n/s]	[n/s]	[n/s]	[n/s]
	192.168.1.19	1902 ms	[n/a]	[n/a]	[n/a]	01:00:5E:7C:...	[n/a]	80,443,8080

5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Hardware Tool	When You Would Use It
Cable Tester	Use it to check whether an Ethernet cable has broken wires, incorrect wiring, or a bad connection.
Crimping Tool	Use it to attach or replace RJ45 connectors when making or repairing Ethernet cables.
Multimeter	Use it to measure voltage, continuity, or resistance when troubleshooting power-related problems in network equipment.